Understood: write-up

Chinmay Raj · Hiya Voice AI Challenge, Accessibility track · Code: https://github.com/chinmayraj28/understood (setup in README.md, AI use in ai.md)

What I built. Understood is a website that takes one sentence of speech, slow, slurred or quiet, and works out what the person said and meant. You can play it a real recording of a person with dysarthria, upload your own, or speak. Every step is drawn on screen as it happens, with real timings, and the transcript is marked word by word against what was really said. Spoken sentences then go to a pretend phone, which acts or asks a short question first.

Who it is for. People with dysarthria: speech made slow, slurred or quiet by cerebral palsy, ALS, Parkinson's, MS or a stroke. On severe dysarthria, off-the-shelf recognisers get more than half the words wrong.

Why voice. Many of these people cannot use their hands well either, so a keyboard is not slower for them, it is impossible or exhausting. Voice is their only way in, and today it works worst for exactly them.

How it works. The sentence is heard several ways at once: by ElevenLabs Scribe, by Gemini after being shown a few recordings of the same person (two Gemini models for a recording), and by Scribe's live transcript. ElevenLabs gives no confidence score, so agreement between them becomes the confidence signal. A phrasebook catches mishearings this person has corrected before. Claude then picks the most likely meaning (Haiku for a spoken sentence, Opus merging the transcripts of a recording), and a fixed rule in code decides whether to act, ask a yes/no question, offer a choice ("Mum or Sam?") or ask once to repeat only the unclear part. **Claude alone never makes it act**: a reading counts only when the recognisers or a twice-confirmed phrasebook entry back it word for word. A wrong action is worse than a slow one, because the person cannot undo it by hand.

In a live check, a synthetic voice said "message Sam running late". Scribe heard "Mum sis, I'm running late" and Gemini timed out. Claude read "message Sam running late" with medium confidence, and the rule asked "Message Sam running late?" instead of guessing.

Does it work? I measured it on TORGO, real recordings of people with dysarthria, setting each speaker up only on their earlier clips. The first version cut words wrong from 34% with Scribe alone to 25%, and acted wrongly on 2% of recordings. On the last day I focused on the transcript itself. I compared 13 recognisers and 5 audio models, and found that Gemini was copying its example clips and that the repair prompt rewrote good transcripts as phone commands. I then rebuilt the listening around two Gemini models and a merging step, and checked the result on held-out clips never used to make a choice. There, on 220 clips, words wrong fell from 40% to 21%, and from 53% to 28% for speakers with dysarthria. That is one dataset and eleven speakers, so I claim only what it shows: the most severe speakers are still often misheard.

How it got here. I started with a Mac app that runs the computer by voice. Feedback from a Hiya engineer pointed me at impaired speech, and I saw it was the bigger gap for the same users, because an assistant that acts for you is useless if it mishears you.

What I learned. First, more evidence is not automatically safer. Adding Gemini let the pipeline act on 39% of clips instead of none, but it was wrong on 7%. The third vote, Claude's repair and the phrasebook brought wrong actions down to 2%. Second, safety rules belong in code. A review found that "Okay, message Sam instead" was being read as a yes to "Message Mum?", so now a yes counts only if it is just a yes, heard after the question finished. Third, never let the model choose who something goes to. A review found that Claude could fill in who a message was for, so "tell her I'm late" could go to whoever it guessed. Now code accepts a name only if it is a known contact and the person actually said it; otherwise the phone asks who. Anything that reaches another person always asks first.

What I would do next. First, accuracy, by learning each voice instead of only prompting. Today nothing is trained: the gains come from choosing and combining recognisers. The next step is training. One route is to fine-tune an open recogniser (such as Whisper) with a small adapter on the Speech Accessibility Project corpus from the University of Illinois: thousands of recordings from people with Parkinson's, ALS, cerebral palsy, Down syndrome and stroke, available to researchers under a data agreement. The other is to personalise for each person. A few minutes of their own speech would train a small adapter for them alone, the approach Google's Project Euphonia showed works for impaired speech. The merge and the ask-first rule would stay on top, and I would test live with people who have dysarthria before claiming anything. Second, speed: stream the audio, and skip the slow listeners when the fast ones agree. Third, make it an API. One call takes audio and returns the transcript, how sure it is, the words in doubt, and whether to act or ask first. The person's voice profile stays on their own device. Any product driven by voice (an assistant, a call-answering agent, a smart home, a car) could put it in front of its own recogniser and become usable by the people it fails today.